Gempack: Its features and capabilities

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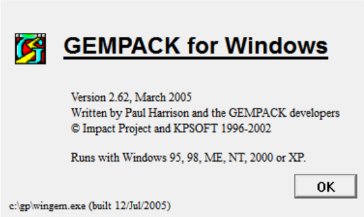
Outline

- Overview of Gempack
- The Gempack Windows programs
- Gempack's way of solving CGE models
- Exercises

Overview



- Gempack (General Equilibrium Modelling Package) A suite of general-purpose economic modelling software.
- Designed for general equilibrium models (unlike GAMS which is designed for any algebraic mathematical modeling)
- Developed by Centre for Policy Studies (CoPS) Monash University, Australia.

Overview (cont.)

- GEMPACK software is being used in over 400 organizations around the world (universities, government departments and private sector firms), about two thirds of which are outside Australia. It is used to implement and solve a number of economic models.
- The software is user-friendly, and fully documented from a user's point of view.
- GEMPACK provides a number of features designed to make this sort of modelling easier.
- GEMPACK runs on a variety of different machines.

Source: https://www.copsmodels.com/gempack.htm

The Gempack Programs provides:

- a simple language in which to describe and document the equations of your economic model;
- a program which converts the equations of your model to a form ready for running simulations with the model;
- options for varying the choice of exogenous and endogenous variables and the variables shocked;
- utility programs to assist in managing the database on which the model is based. The data can be inspected, modified, converted to spreadsheets or moved to different machines (including those with different operating systems).

Gempak Windows program

1. WinGEM $\Rightarrow$ Windows interface to GEMPACK
2. TABmate $\Rightarrow$ Windows text editor for developing TABLO Input files
3. ViewHAR $\Rightarrow$ Windows program for looking at data in a Header Array file
4. ViewSOL $\Rightarrow$ Windows program for looking at Solution files
5. AnalyseGE $\Rightarrow$ Windows program assisting modellers to analyse their results
6. RunGEM $\Rightarrow$ Windows program for automating simulations with models
7. RunDynam $\Rightarrow$ Windows interface for recursive dynamic models
8. Charter $\Rightarrow$ Windows program for drawing graphs

- *We are going to use mostly 1-4! We will use 5 and 10 sometimes.*

Four most often used Gempack Windows program in this course

1. WinGEM ⇒ Windows interface to GEMPACK
2. TABmate ⇒ Windows text editor for developing TABLO Input files
3. ViewHAR ⇒ Windows program for looking at data in a Header Array file
4. ViewSOL ⇒ Windows program for looking at Solution files

Gempack's way of solving CGE models

- CGE model is simply a system of non-linear equations of n variables and n equations. GEMPACK solves the system.
- How? Simply through Matrix Inversion.
- In GEMPACK we work with *linearized equations*, the CGE model become a system of linear equations.
- But, that's not the only thing GEMPACK do. Gempack correct the linearization error such that the solution is as if solved through non-linear computation.

GEMPACK Solution Method

Refer to previous module, the solution for K_1, L_1, and X_1 are,

$$\begin{aligned} K_1 &= \frac{\alpha\gamma}{\beta(1-\gamma)+\alpha\gamma} K_S \\ L_1 &= \frac{(1-\alpha)\gamma}{(1-\beta)(1-\gamma)+(1-\alpha)\gamma} L_S \\ X_1 &= K_1^{\alpha} L_1^{1-\alpha} \end{aligned}$$

Substituting K_1 and L_1 to X_1, $X_1 = AL_S^{1-\alpha}$ where:

$$A = \left(\frac{\alpha\gamma}{\beta(1-\gamma)+\alpha\gamma} K_S\right)^{\alpha} \left(\frac{(1-\alpha)\gamma}{(1-\beta)(1-\gamma)+(1-\alpha)\gamma}\right)^{1-\alpha}$$

GEMPACK Solution Method (cont.)

From our database, $K_{S0} = 891.72$, $L_{S0} = 408.18$, $\alpha = 0.804, \beta = 0.643$, $\gamma = 0.267$. Then $X_1 = 64.88 L_S^{0.196}$. Given K_S, X_1 is a non-linear function of L_S. If L_S change 200% from $L_{S0} = 408.18$ to $L_{S1} = 1224.55$, X_1 changed from

$$
\begin{aligned}
X_1^0 &= 64.88 \cdot \left(408.18^{0.196}\right) = 211.22, \text{ to} \\
X_1^1 &= 64.88 \cdot \left(1224.55^{0.196}\right) = 262.06
\end{aligned}
$$

$$
\text{or by } 100 \cdot \left(\frac{X_1^1}{X_1^0} - 1\right) = 100 \cdot \left(\frac{262.06}{211.22} - 1\right) = 24.07\%
$$

This is the exact (non-linear) solution of percentage change in X_1 with respect to change in L_S by 200%.

GEMPACK Solution Method (cont.)

The linearized (percentage change) form of

$$X_1 = AL_S^{1-\alpha} \text{ is}$$

$$x_1 = (1-\alpha)\, l_S \text{ or}$$

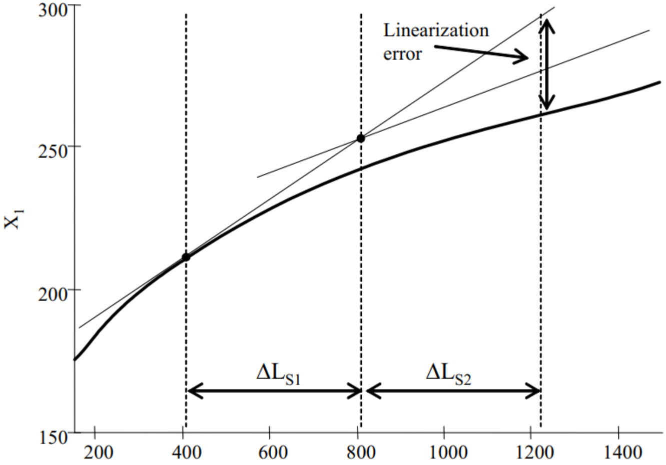
$$x_1 = 0.196 \cdot l_S$$

therefore if the change in L_S is 200%, or $l_S = 200$, then

$$x_1 = 0.196 \cdot 200 = 39.27\%$$

which suffer from linearization error of $39.27 - 24.07 = 15.19\%$
Linearized solution only accurate for small changes.

GEMPACK Correcting linearisation error



Correcting Linearization Error (cont.)

The calculus (linearized) aproximation of change in X_1 with respect to change in L_S is

$$\begin{aligned} \Delta X_1 &= \frac{\partial X_1}{\partial L_S} \Delta L_S \\ \Delta X_1 &= f(L_S) \cdot \Delta L_S \end{aligned}$$

with $X_1 = AL_S^{1-\alpha}$

$$f(L_S) = (1-\alpha) AL_S^{-\alpha}$$

Divide this shock into two parts, $\Delta LS^1 = \frac{1}{2} \cdot 816.36 = 408.18$. First we calculate the derivative $f\left(L_{S0}^1 = 408.18\right) = (1 - 0.196) \cdot 64.88 \cdot 408.18^{-0.804} = 0.102$, then

$$\Delta X_1^1 = f\left(L_{S0}^1\right) \cdot \Delta L_S^1 = 0.102 \cdot 408.18 = 41.47$$

Correcting Linearization Error (cont.)

Now, let's update the derivative
$f\left(L_{S0}^2 = L_{S0}^1 + \Delta LS^1\right) = f\left(L_{S0}^2 = 816.36\right) = 0.196 \cdot 64.88 \cdot 816.36^{-0.804} = 0.058.$

$$\Delta X_1^2 = f\left(L_{S0}^2\right) \cdot \Delta L_S^2 = 0.058 \cdot 408.18 = 23.76$$

And finally, combine the two parts

$$\Delta X_1 = \Delta X_1^1 + \Delta X_1^2 = 41.47 + 23.76 = 65.23$$

$$\text{In percentage change } x_1 = 100 \cdot \frac{\Delta X_1}{X_{10}} = 100 \cdot \frac{65.23}{211.22} = 30.88\%$$

The linearization error drop from 15.19% into only $30.88 - 24.07 = 6.81\%$. More partion, more accurate.In fact, GEMPACK can produce solution with a very high accuracy by extrapolation.

Excercise: Illustration of GEMPACK solution method

1. Open exercise1.TAB, modify it to represent Cobb-Douglass production function!
2. Open exercise1b.CMF, add shock statement of increase in labor supply by 200%.
3. In this file, notice that solution method is Johansen, this is single step linearization.
4. Do the simulation, How much change in X_1? is it exact, or by 24.07%? why?
5. Gempack produces a warning. Investigate what the warning is.
6. Now change the solution method using this one:
   Method = Gragg; Steps = 4 6 8;
7. Now, how much change in X_1? is it exact now? How come? What GEMPACK really do?

What does GEMPACK do?

With GRAGG steps 4 6 8

- Note, the exact solution is $x_1 = 24.07\%$, the linearized (johansen) solution is $x_1 = 39.27\%$.
- First step: GEMPACK divide the shock into 4 partitions, the result is $x_1 = 24.49\%$.
- Second step: GEMPACK divide the shock into more partitions 6, the result is $x_1 = 24.26\%$. This result is better than first one.
- Last step: GEMPACK divide the shock into 8 partitions, the result is $x_1 = 24.18\%$. This result is even better.
- Here, GEMPACK actually has solve using Johansen single step how many times? $4 + 6 + 8 = 18$ times.
- Then GEMPACK see the series $24.49, 24.26, 24.18$, and make extrapolation for the exact solution, 24.07.